

Electromagnetic Waves

Static Electric Fields



Fundamental laws:

- Coulomb's Law
- Gauss's Law

(both these laws are based on experimental studies and are interdependent)

Fundamental Postulates of Electrostatics in Free Space



Electric field intensity

$$\mathbf{E} = \lim_{q \rightarrow 0} \frac{\mathbf{F}}{q} \quad (\text{V/m}).$$

Inverse Relation:

$$\mathbf{F} = q\mathbf{E} \quad (\text{N}).$$

Fundamental Postulates:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \times \mathbf{E} = 0$$

Fundamental Postulates of Electrostatics in Free Space



Fundamental Postulates:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \times \mathbf{E} = 0$$

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

Gauss's Law

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0$$

Total outward flux of the E-field over any closed surface in free space is equal to the total (net) charge enclosed in the surface divided by ϵ_0 .

The scalar line integral of the static electric field intensity around any closed path vanishes

Fundamental Postulates of Electrostatics in Free Space



Summary

Postulates of Electrostatics in Free Space	
Differential Form	Integral Form
$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$	$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$
$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0$

Fundamental Postulates of Magnetostatics in Free Space



A magnetostatic field is produced by a constant current flow.

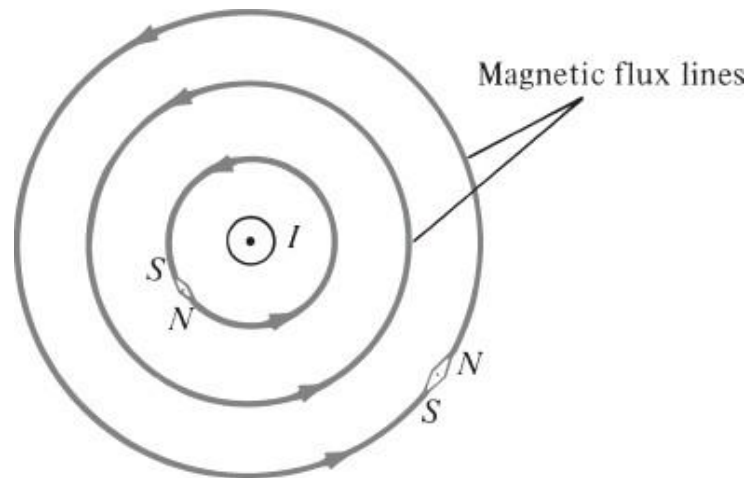
1. Divergence of $\mathbf{B}$

2. Curl of $\mathbf{B}$

Fundamental Postulates of Magnetostatics in Free Space



$$\psi = \int_s B \cdot dS$$



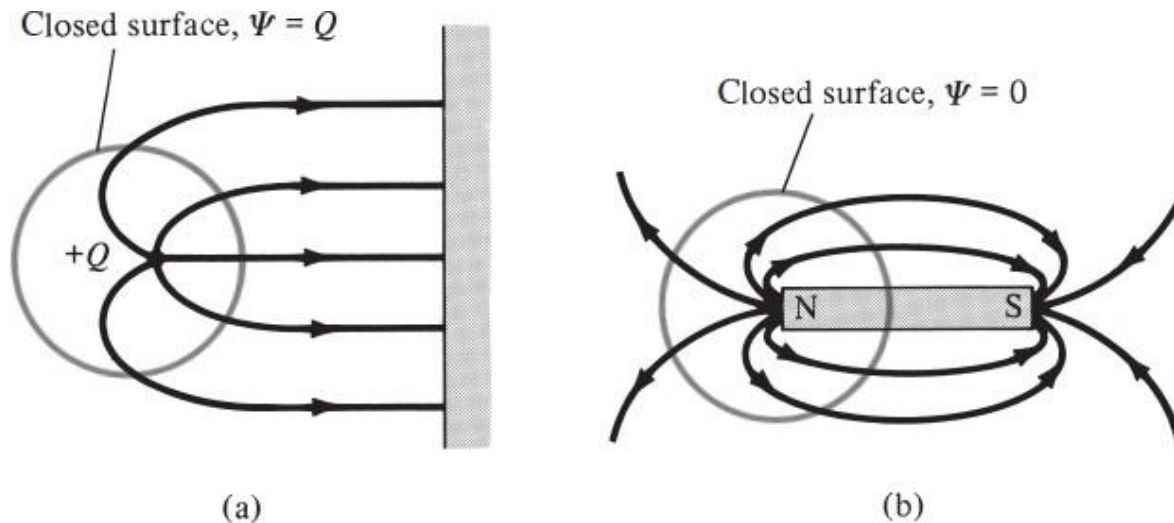


Figure 6.17 Flux leaving a closed surface due to (a) isolated electric charge $\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q$, (b) magnetic charge, $\Psi = \oint_S \mathbf{B} \cdot d\mathbf{S} = 0$.

$$\oint_c \vec{B} \cdot d\vec{S} = 0$$
$$\vec{\nabla} \cdot \vec{B} = 0$$

Gauss's Law for magnetic fields

It is not possible to have isolated magnetic poles

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

Tell us try to have its integral form



Ampere's Circuital Law

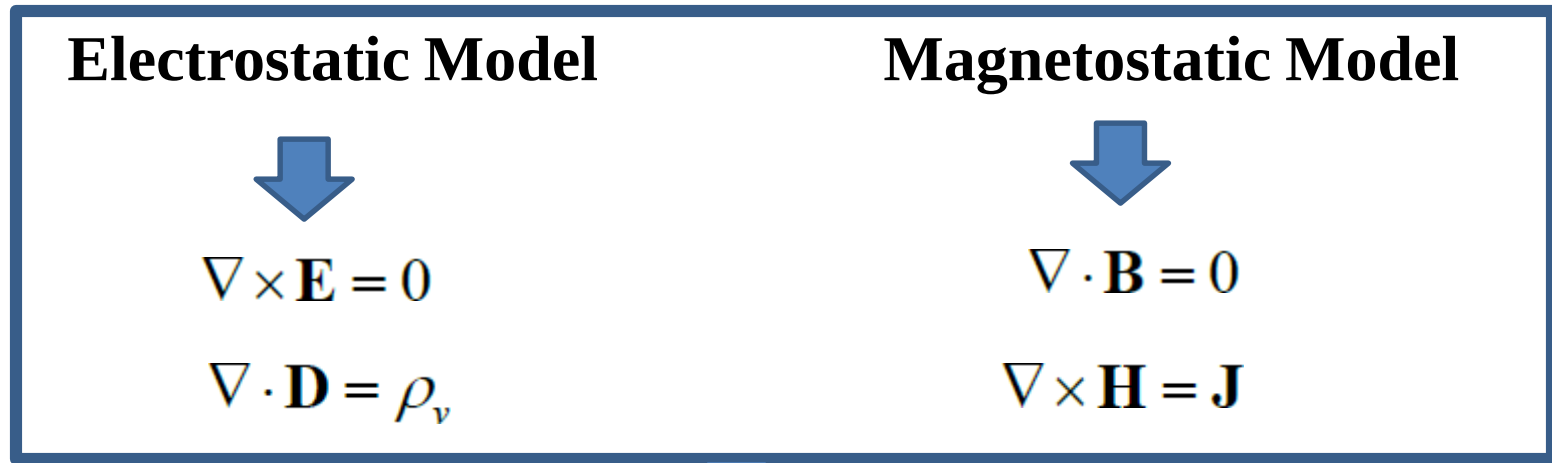
Ampere's circuital law states that the line integral of $\mathbf{H}$ around a closed path is the same as the net current enclosed by the path.

Magnetostatic Field is not conservative

Maxwell's Equation for **Static** electric field and **steady** magnetic field

Differential (or Point) Form	Integral Form	Remarks
1. $\nabla \cdot \mathbf{D} = \rho_v$	$\oint_s \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv$	Gauss's Law
2. $\nabla \cdot \mathbf{B} = 0$	$\oint_s \mathbf{B} \cdot d\mathbf{S} = 0$	Nonexistence of magnetic monopole
3. $\nabla \times \mathbf{E} = 0$	$\oint_s \mathbf{E} \cdot d\mathbf{l} = 0$	Conservative nature of electrostatic field.
4. $\nabla \times \mathbf{H} = \mathbf{J}$	$\oint_s \mathbf{H} \cdot d\mathbf{l} = \int_s \mathbf{J} \cdot d\mathbf{S}$	Ampere's Law

Maxwell's equations (Time Varying)



Static EM fields, electric and magnetic fields are **independent** of each other

However

Dynamic EM fields, electric and magnetic fields are **interdependent** of each other



Electromagnetic Model

Faraday's Law of Electromagnetic Induction



Experimentally discovered: A current was induced in a conducting loop when the magnetic flux linking the loop changed.

Fundamental Postulate for Electromagnetic Induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

It says that the electric field intensity in a region of time-varying magnetic flux density is therefore non-conservative and cannot be expressed as the gradient of a scalar potential.

Maxwell's Equations



$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J}$$

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \neq 0$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d$$

Concept of Displacement Current:

Maxwell's Equations in Time-varying Fields



<u>Differential</u>	<u>Integral</u>	<u>Remarks</u>
1 $\nabla \cdot D = \rho_v$	$\oint_S D \cdot dS = \int_V \rho_v dv$	<u>Gauss' Law</u>
2 $\nabla \cdot B = 0$	$\oint_S B \cdot dS = 0$	<u>Nonexistence of isolated charge</u>
3 $\nabla \times E = -\frac{\partial B}{\partial t}$	$\oint_L E \cdot dl = -\frac{\partial}{\partial t} \int_S B \cdot dS$	<u>Faraday's Law</u>
4 $\nabla \times H = J + \frac{\partial D}{\partial t}$	$\oint_L H \cdot dl = \int_S \left(J + \frac{\partial D}{\partial t} \right) \cdot dS$	<u>Modified Ampere's law</u>

In 1 and 2, S is a closed surface enclosing the volume V

In 3 and 4, L is a closed path that bounds the surface S

Concept of Displacement Current:

Maxwell's Equations in Time-varying Fields



$$\nabla \times \vec{H} = \vec{J} + \vec{J}_D$$

$$\vec{D} = \epsilon \vec{E}$$

$$\vec{E} = E_0 e^{j\omega t}$$

$$\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$$

$$\frac{\partial \vec{E}}{\partial t} = j\omega \vec{E}$$

$$\nabla \times \vec{H} = \sigma \vec{E} + j\omega \epsilon \vec{E}$$

↓ Dielectric

↓ Conductor

$$\left[f = \frac{\sigma}{2\pi \epsilon_0 \epsilon_r} \right]$$

conduction current = Displacement current

Uniform Plane Wave



free space charge free region $\rho = 0$
 $\rho = 0$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \rightarrow \nabla \times (\nabla \times \vec{E}) = -\frac{\partial \mu (\nabla \times \vec{H})}{\partial t}$$
$$\nabla \times \vec{H} = \frac{\partial \vec{B}}{\partial t}$$
$$\nabla^2 \vec{E} = -\mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

$$\nabla^2 \vec{E} = \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

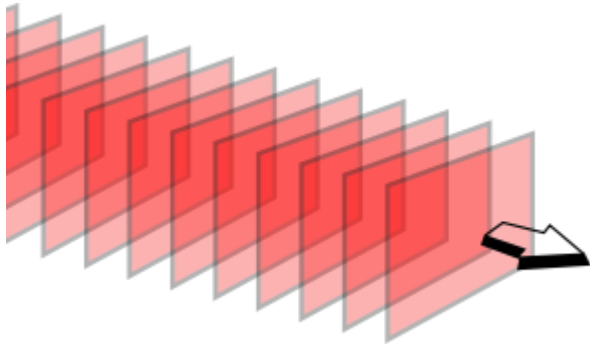
$$\nabla^2 \vec{H} = \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

} Helmholtz's
Wave eqⁿ

Uniform Plane Wave



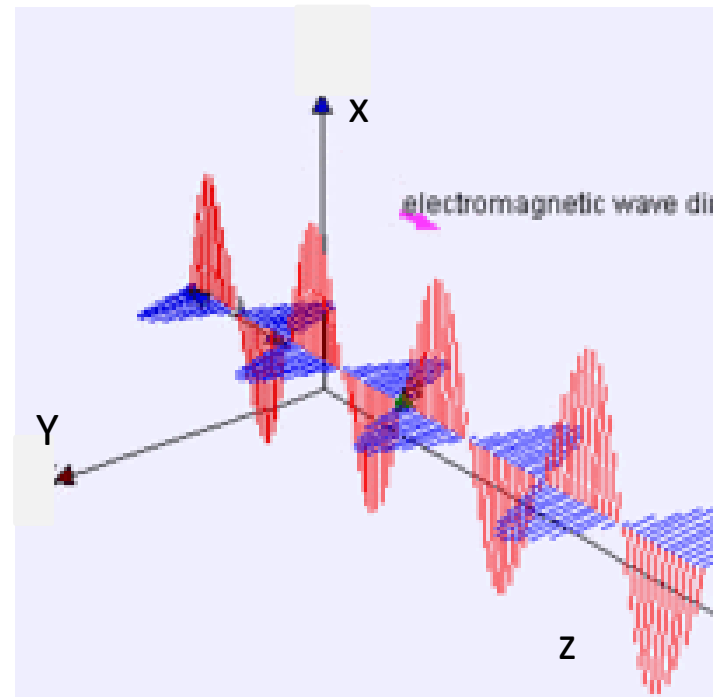
Electric Field (and also $\mathbf{H}$) assumed to have same direction, same magnitude, and same phase in infinite planes perpendicular to the direction of propagation.



The wavefronts of a plane wave traveling in space

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0,$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \cong 3 \times 10^8 \text{ (m/s)} = 300 \text{ (Mm/s)}$$



Plane Waves in Lossless Media



$$\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = 0,$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k_0^2 \right) E_x = 0.$$

$$\frac{d^2 E_x}{dz^2} + k_0^2 E_x = 0,$$

$$\begin{aligned} E_x(z) &= E_x^+(z) + E_x^-(z) \\ &= E_0^+ e^{-jk_0 z} + E_0^- e^{jk_0 z} \end{aligned}$$

$$\begin{aligned} E_x^+(z, t) &= \Re [E_x^+(z) e^{j\omega t}] \\ &= \Re [E_0^+ e^{j(\omega t - k_0 z)}] \\ &= E_0^+ \cos(\omega t - k_0 z) \quad (\text{V/m}). \end{aligned}$$

$$k_0 = \frac{2\pi}{\lambda_0} \quad (\text{rad/m}),$$

$\omega t - k_0 z = \text{A constant phase,}$

$$\frac{dz}{dt} = \frac{\omega}{k_0} = c.$$

Plane Waves in Lossless Media

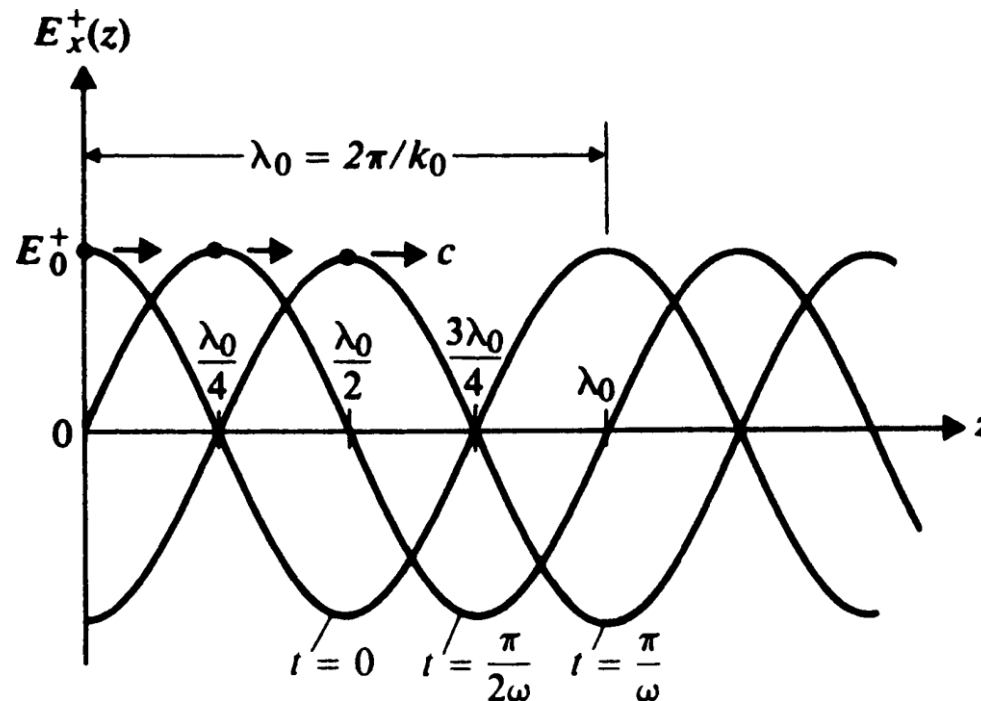


Considering the source free wave equation for free space:

$$\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = 0,$$

$$k_0 = \omega \sqrt{\mu_0 \epsilon_0} = \frac{\omega}{c} \quad (\text{rad/m}).$$

Traveling Wave



Relationship between E and H



$$\nabla \times \vec{E} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ 0 & 0 & \frac{\partial}{\partial z} \\ \vec{E}_x(z) & 0 & 0 \end{vmatrix}$$

$$\vec{E} = (E^+ e^{-kz} + E^- e^{+kz}) \hat{a}_z$$

$$\vec{H} = (\mu^+ e^{-kz} + \mu^- e^{+kz}) \hat{a}_y$$

$$= -j\omega\mu_0(\vec{a}_x H_x + \vec{a}_y H_y + \vec{a}_z H_z),$$

$$\left[\frac{E^+}{\mu^+} = \sqrt{\frac{\mu_0}{\epsilon_0}} = -\frac{E^-}{\mu^-} \right]$$

$$\begin{aligned} H_x &= 0 \\ H_y &= \frac{1}{-j\omega\mu_0} \frac{\partial E_x(z)}{\partial z} \\ H_z &= 0 \end{aligned}$$

$$\frac{1}{j\omega\mu_0} \frac{\partial E_z}{\partial z} = \left[\frac{k E^+ e^{-kz}}{j\omega\mu_0} - \frac{k E^- e^{+kz}}{j\omega\mu_0} \right]$$

$$\Rightarrow H_y = \left[\sqrt{\frac{\epsilon_0}{\mu_0}} E^+ e^{-kz} - \sqrt{\frac{\epsilon_0}{\mu_0}} E^- e^{+kz} \right]$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \cong 120\pi \cong 377 \quad (\Omega),$$

Intrinsic impedance of free space

$$\left[\frac{E^+}{\mu^+} = -\frac{E^-}{\mu^-} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi \right]$$

Source-Free Fields in Simple Media



$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H},$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E},$$

$$\nabla \cdot \mathbf{E} = 0,$$

$$\nabla \cdot \mathbf{H} = 0.$$

$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0$$

$$\nabla^2 \mathbf{H} + k^2 \mathbf{H} = 0,$$

Homogeneous vector Helmholtz's Equations

If the simple medium is **conducting**, a current $\mathbf{J}$ will flow



Concept of complex permittivity and loss tangent

Source-Free Simple Conducting Media



If the simple medium is **conducting**, a current $\mathbf{J}$ will flow

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H},$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E},$$

$$\nabla \cdot \mathbf{E} = 0,$$

$$\nabla \cdot \mathbf{H} = 0.$$

$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0$$

$$\nabla^2 \mathbf{H} + k^2 \mathbf{H} = 0,$$

$$\times \longrightarrow \nabla \times \vec{H} = \sigma \vec{E} + j\omega\epsilon \vec{E}$$

$$\nabla \times \vec{H} = (\sigma + j\omega\epsilon) \vec{E}$$

$$\nabla \times \vec{H} = j\omega\epsilon \left[1 - \frac{j\sigma}{\omega\epsilon} \right] \vec{E}$$

$\sigma \neq 0$ system

$$\nabla \times \vec{H} = j\omega\epsilon_0\epsilon_r \left(1 - \frac{j\sigma}{\omega\epsilon_0\epsilon_r} \right) \vec{E} \text{ is Lossy}$$

Source-Free Simple Conducting Media



Concept of complex permittivity and loss tangent

$$\epsilon_c = \epsilon' - j\epsilon'' \quad (\text{F/m}),$$

real complex

$$\tan \delta_c = \frac{\epsilon''}{\epsilon'} \approx \frac{\sigma}{\omega \epsilon'}$$

loss tangent

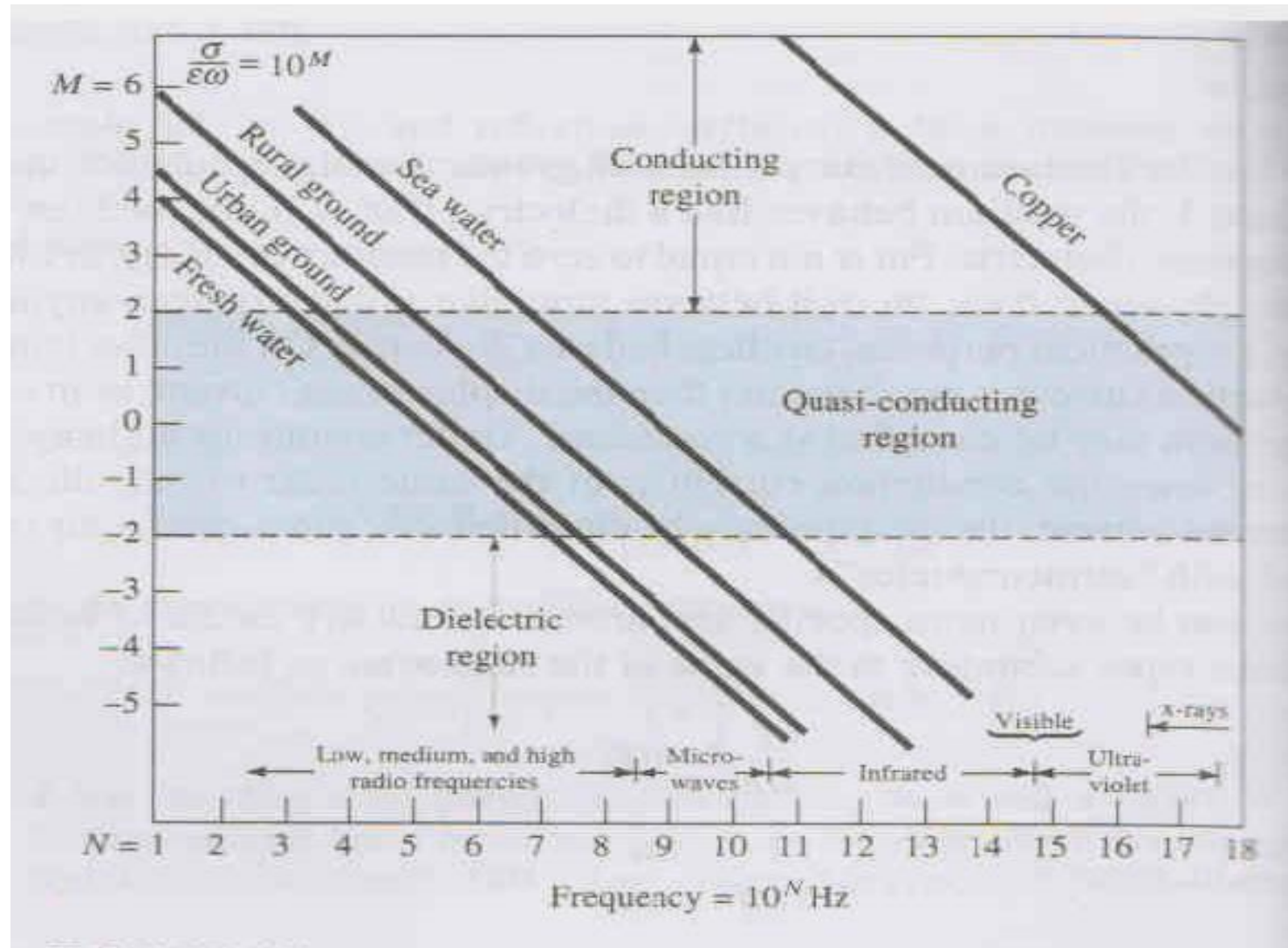
$$K = j\omega \sqrt{\mu \epsilon} = j\omega \sqrt{\mu_0 \epsilon_0 \mu_r} \sqrt{\epsilon_r \left(1 - j \frac{\sigma}{\omega \epsilon_0 \epsilon_r} \right)}$$

- Importance of loss tangent:

- Measure of the power loss in the medium
- Also used in deciding the type of medium (at operating frequency)

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}} \sqrt{\frac{1}{\epsilon_r \left(1 - j \frac{\sigma}{\omega \epsilon_0 \epsilon_r} \right)}}$$

Medium behavior at different frequencies



Types of Medium



$$\sigma \neq 0$$



Lossy Medium

Low-loss Dielectrics

σ is small

Good conductors

σ is very large

$$\sigma = 0$$



Lossless Medium

$$\epsilon = \epsilon_r \epsilon_0$$

$$\mu = \mu_r \mu_0$$

1. Free space ($\sigma = 0, \epsilon = \epsilon_0, \mu = \mu_0$)
2. Lossless dielectrics ($\sigma = 0, \epsilon = \epsilon_r \epsilon_0, \mu = \mu_r \mu_0$)
3. Lossy dielectrics ($\sigma \neq 0, \epsilon = \epsilon_r \epsilon_0, \mu = \mu_r \mu_0, \sigma \ll \omega \epsilon$)
4. Good conductors ($\sigma \simeq \infty, \epsilon = \epsilon_0, \mu = \mu_r \mu_0$, or $\sigma \gg \omega \epsilon$)

Plane Waves in Lossy Media (σ is finite)



Complex Permittivity

$$\epsilon_c = \epsilon \left[1 - j \left(\frac{\sigma}{\omega \epsilon} \right) \right] = \epsilon' - j\epsilon'' = |\epsilon_c| e^{-j\theta}$$

$$\epsilon' = \epsilon \quad \epsilon'' = \frac{\sigma}{\omega}$$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta$$

$$\gamma = j\omega\sqrt{\mu\epsilon} \sqrt{1 - j \frac{\sigma}{\omega\epsilon}}$$

$$\gamma = \alpha + j\beta = j\omega\sqrt{\mu\epsilon'} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{1/2},$$

Intrinsic Impedance



$$\nabla \times \mathbf{E} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & 0 & \frac{\partial}{\partial z} \\ E_x^+(z) & 0 & 0 \end{vmatrix}$$

$$\begin{aligned} H_x^+ &= 0 \\ H_y^+ &= \frac{1}{-j\omega\mu_0} \frac{\partial E_x^+(z)}{\partial z} \\ H_z^+ &= 0 \end{aligned}$$

$$\nabla^2 \mathbf{E}_s - \gamma^2 \mathbf{E}_s = 0$$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = |\eta| \angle \theta_\eta = |\eta| e^{j\theta_\eta}$$

E and H relationship



If the electric field propagate in +z direction and has component x, the equation of the wave is given by :

$$\bar{E}(z, t) = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \hat{a}_x$$

And the magnetic field :

$$\bar{H}(z, t) = H_0 e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \hat{a}_y$$

$$H_0 = \frac{E_0}{|\eta|}$$

Conclusions that can be made for the wave propagating in *lossy dielectrics material* :

- (i)** E and H fields amplitude will be **attenuated by** $e^{-\alpha z}$
- (ii)** E **leading** H by θ_η

Lossless (Perfect Dielectrics)



Characteristics: $\sigma = 0, \epsilon = \epsilon_0 \epsilon_r, \mu = \mu_0 \mu_r$

$$\alpha = 0, \quad \beta = \omega \sqrt{\mu \epsilon}$$

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu \epsilon}}, \quad \lambda = \frac{2\pi}{\beta}$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \angle 0^\circ$$

The zero angle means that E and H fields are in phase at each fixed location.

Free Space



Characteristics: $\sigma = 0, \epsilon = \epsilon_0, \mu = \mu_0$

$$\alpha = 0, \quad \beta = \omega \sqrt{\mu_0 \epsilon_0}$$

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}, \quad \lambda = \frac{2\pi}{\beta}$$

$$\eta = \sqrt{\frac{\mu_0}{\epsilon_0}} \angle 0^\circ = 120\pi = 377\Omega$$

The zero angle means that E and H fields are in phase at each fixed location.

Good Conductors



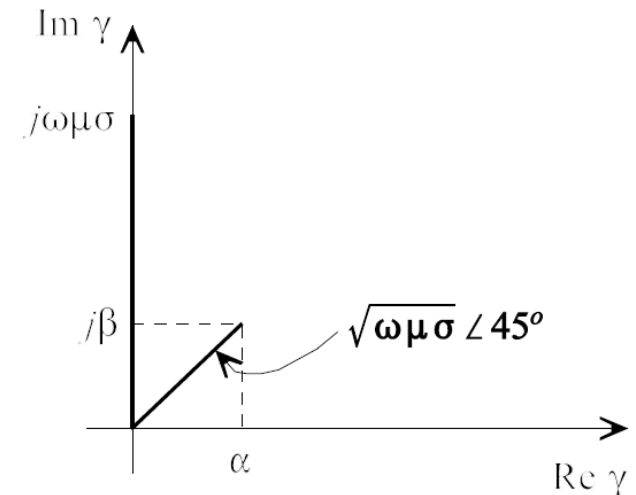
In conductors : $\sigma \gg \omega \epsilon$ or $\frac{\sigma}{\omega \epsilon} \gg 1$

With the characteristics : $\sigma \sim \infty, \epsilon = \epsilon_r \epsilon_0, \mu = \mu_0 \mu_r$

$$\alpha = \beta = \sqrt{\frac{\omega \mu \sigma}{2}} = \sqrt{\pi f \mu \sigma}$$

$$\eta = \sqrt{\frac{\omega \mu}{\sigma}} \angle 45^\circ$$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$



The field equations for E and H obtained :

$$\bar{E} = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \hat{a}_x$$

$$\bar{H} = \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - 45^\circ) \hat{a}_y$$

$$v = \frac{\omega}{\beta} = \sqrt{\frac{2\omega}{\mu\sigma}}$$

Skin depth



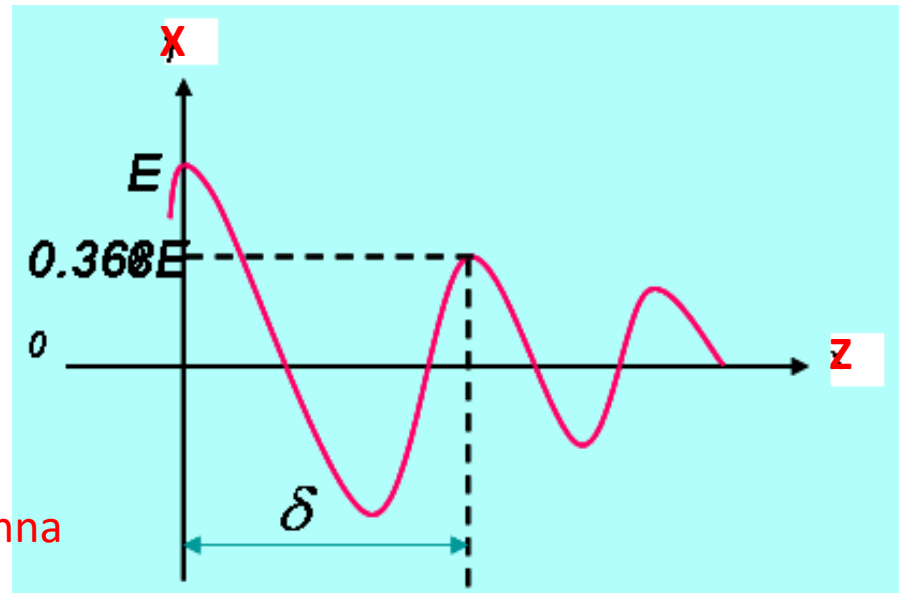
It is seen that in conductors $\bar{E}$ and $\bar{H}$ waves are attenuated by $e^{-\alpha z}$

*From the diagram δ is referred to as the **skin depth**. It refers to the amplitude of the wave propagate to a conducting media is reduced to e^{-1} or 37% from its initial value. also known as **1/e depth of penetration***

In a distance :

$$E_0 e^{-\alpha \delta} = E_0 e^{-1}$$

$$\delta = 1 / \alpha = \frac{1}{\sqrt{\pi f \mu \sigma}}$$



Communication application : Hollow antenna

The phenomenon whereby field intensity in a conductor rapidly decreases is known as **skin effect**.

Physical significance of skin depth



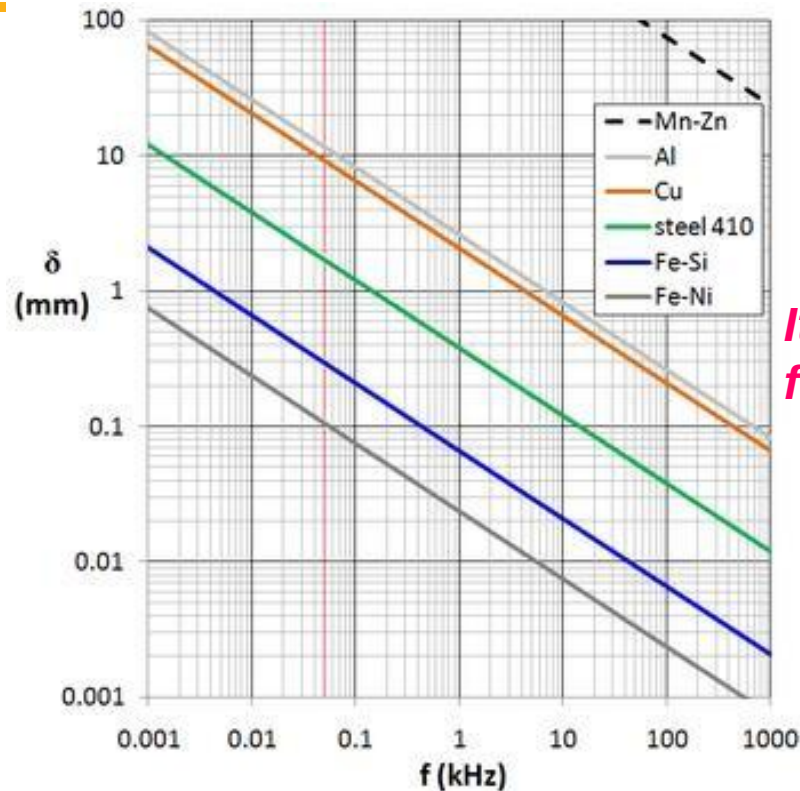
Physical significance of skin depth

It shows that any electromagnetic wave with high frequency cannot propagate through the conducting media. The large value of skin depth signifies the less attenuation of electromagnetic waves in any medium.

$$\delta = 1 / \alpha = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

A high-frequency field attenuates as it penetrates a conductor in a shorter distance than a low-frequency field **low pass filter**

Skin depth



$$\delta \propto \frac{1}{\sqrt{f}}$$

It can be seen that at higher frequencies δ is decreasing.

Skin Depths, δ in (mm), of Various Materials

Material	σ (S/m)	$f = 60$ (Hz)	1 (MHz)	1 (GHz)
Silver	6.17×10^7	8.27 (mm)	0.064 (mm)	0.0020 (mm)
Copper	5.80×10^7	8.53	0.066	0.0021
Gold	4.10×10^7	10.14	0.079	0.0025
Aluminum	3.54×10^7	10.92	0.084	0.0027
Iron ($\mu_r \cong 10^3$)	1.00×10^7	0.65	0.005	0.00016
Seawater	4	32 (m)	0.25 (m)	†

EM Power and Poynting Vector



$$\left[\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \right]$$

$$\left[\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \right]$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H})$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = - \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \vec{J} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$$

EM Power and Poynting Vector



Energy is transported through space to distant receiving points by EM waves.

$$\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = \mathbf{H} \cdot \frac{\partial(\mu \mathbf{H})}{\partial t} = \frac{1}{2} \frac{\partial(\mu \mathbf{H} \cdot \mathbf{H})}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \mu H^2 \right),$$

$$\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} = \mathbf{E} \cdot \frac{\partial(\epsilon \mathbf{E})}{\partial t} = \frac{1}{2} \frac{\partial(\epsilon \mathbf{E} \cdot \mathbf{E})}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 \right),$$

$$\mathbf{E} \cdot \mathbf{J} = \mathbf{E} \cdot (\sigma \mathbf{E}) = \sigma E^2.$$

$$\int \nabla \cdot (\mathbf{E} \times \mathbf{H}) = - \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) - \sigma E^2, \quad \downarrow \checkmark$$

$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{s} = - \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dv - \int_V \sigma E^2 dv,$$

Poynting Vector (power surface density in W/m²)

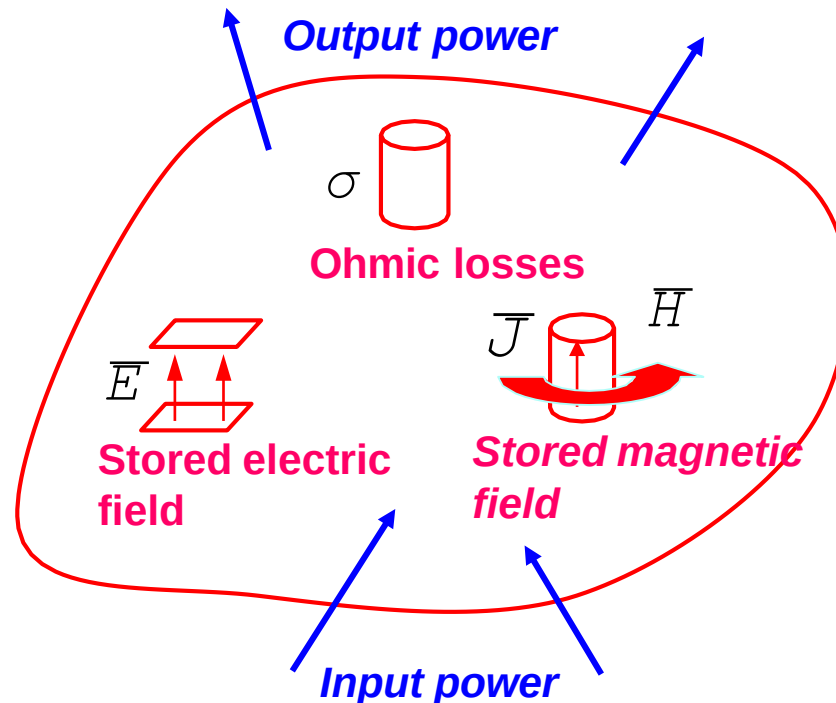
$$\mathbf{P} = \vec{E} \times \vec{H}$$

EM Power and Poynting Vector



Poynting theorem states that the **total power flow leaving the volume** is equal to **the decrease of the energy densities of energy stored in the electric and magnetic fields** and **the dissipated ohmic power**.

The theorem can be explained as shown in the diagram below :



Instantaneous and Average Power Densities



*Given for **lossless dielectric**, the electric and magnetic fields are :*

$$\begin{aligned}\bar{E} &= E_0 \cos(\omega t - \beta z) \hat{a}_x \\ \bar{H} &= \frac{E_0}{|\eta|} \cos(\omega t - \beta z) \hat{a}_y\end{aligned}$$

The Poynting vector becomes:

$$\mathbf{P} = \vec{E} \times \vec{H} \quad \text{W / m}^2$$

The Instantaneous expression of Poynting vector becomes:

$$\mathbf{P}(z, t) = \frac{E_0^2}{|\eta|} \cos^2(\omega t - \beta z) \hat{a}_z$$

Instantaneous and Average Power Densities



To find average power density :

Integrate Poynting vector and divide with interval $T = 1/f$:

*Average power
through area A :*

$$\mathbf{P}_{av} = \frac{1}{2} \frac{E_0^2}{|\eta|} A \quad (W)$$

Instantaneous and Average Power Densities



Given for lossy dielectric, the electric and magnetic fields are :

$$\vec{E} = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \hat{a}_x$$

$$\vec{H} = \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \hat{a}_y$$

$$\mathbf{P} = \vec{E} \times \vec{H} \quad \text{W / m}^2$$

The Instantaneous expression of Poynting vector becomes:

$$\mathbf{P}(z, t) = \frac{E_0^2}{|\eta|} e^{-2\alpha z} \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta) \hat{a}_z$$

Average power :

$$\mathbf{P}_{av} = \frac{1}{2} \frac{E_0^2}{|\eta|} e^{-2\alpha z} \cos \theta_\eta$$

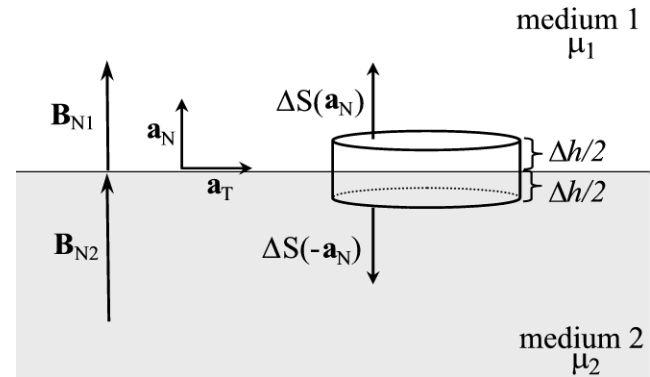
Magnetostatic boundary conditions



Gauss's Law for Magnetostatic fields:

$$\oint \mathbf{B} \cdot d\mathbf{S} = 0$$

$$\begin{aligned} \oint \mathbf{B} \cdot d\mathbf{S} &= \int B_{N1} \mathbf{a}_N \cdot dS \mathbf{a}_N + \int B_{N2} \mathbf{a}_N \cdot dS (-\mathbf{a}_N) \\ &= (B_{N1} - B_{N2}) \Delta S = 0. \end{aligned}$$



Normal Boundary Condition:

$$\boxed{B_{N1} = B_{N2}}.$$

The normal component of the magnetic flux density must be continuous across the boundary.

Important Note:

$$B_{N1} = B_{N2} \longrightarrow \mu_o \mu_1 H_{N1} = \mu_o \mu_2 H_{N2}$$

Therefore, we can say that $H_{N1} \neq H_{N2}$

The normal component of the magnetic field intensity is not continuous across the boundary (but the magnetic flux density is continuous).

Magnetostatic boundary conditions



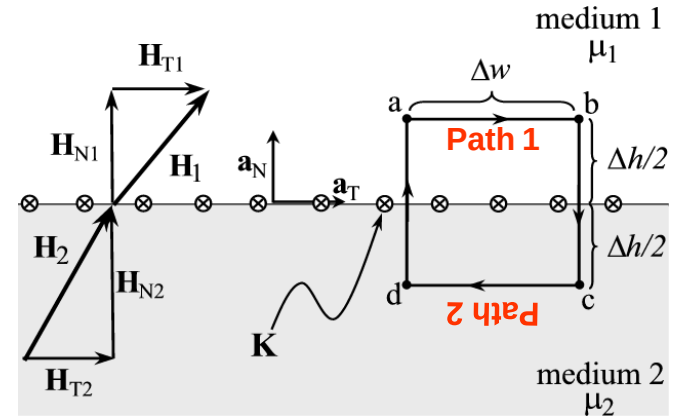
Laws used: Ampere's circuital law and Gauss's law for Magnetostatics

Ampere's circuit law:

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{enc}$$

The current enclosed by the path is

$$I_{enc} = \int K dW = K \Delta w.$$



The circulation of $\mathbf{H}$ is made up of four line integrals:

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_a^b + \int_b^c + \int_c^d + \int_d^a (\mathbf{H} \cdot d\mathbf{L}) = K \Delta w.$$

Path 1:
$$\int_a^b \mathbf{H} \cdot d\mathbf{L} = \int_0^{\Delta w} H_{T1} \mathbf{a}_T \cdot d\mathbf{L} \mathbf{a}_T = H_{T1} \Delta w.$$

Magnetostatic boundary conditions



Path 2:
$$\int_c^d \mathbf{H} \cdot d\mathbf{L} = \int_{\Delta w}^0 H_{T2} \mathbf{a}_T \cdot dL \mathbf{a}_T = -H_{T2} \Delta w.$$

(Path 1 + Path 2), gives

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{enc}$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = (H_{T1} - H_{T2}) \Delta w$$

$$I_{enc} = \int K dW = K \Delta w$$

Tangential Boundary Condition:

$$\boxed{H_{T1} - H_{T2} = K}$$

Magnetostatic boundary conditions



Special Case: If the surface current density $K = 0$, we get

$$H_{T1} - H_{T2} = K \xrightarrow{\text{If } K = 0} H_{T1} = H_{T2}$$

The tangential magnetic field intensity is continuous across the boundary when the surface current density is zero.

Important Note:

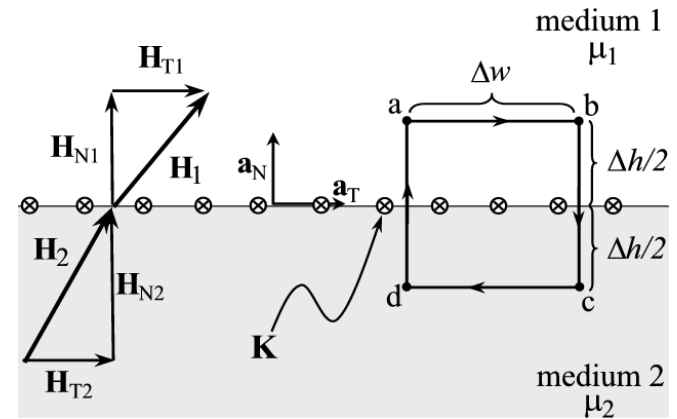
We know that $\mathbf{B} = \mu_o \mu_r \mathbf{H}$ $\mathbf{H} = \frac{\mathbf{B}}{\mu_o \mu_r}$

Using the above relation, we obtain

$$H_{T1} = H_{T2} \xrightarrow{\hspace{1cm}} \frac{B_{T1}}{\mu_o \mu_1} = \frac{B_{T2}}{\mu_o \mu_2}$$

Therefore, we can say that

$$B_{T1} \neq B_{T2}$$

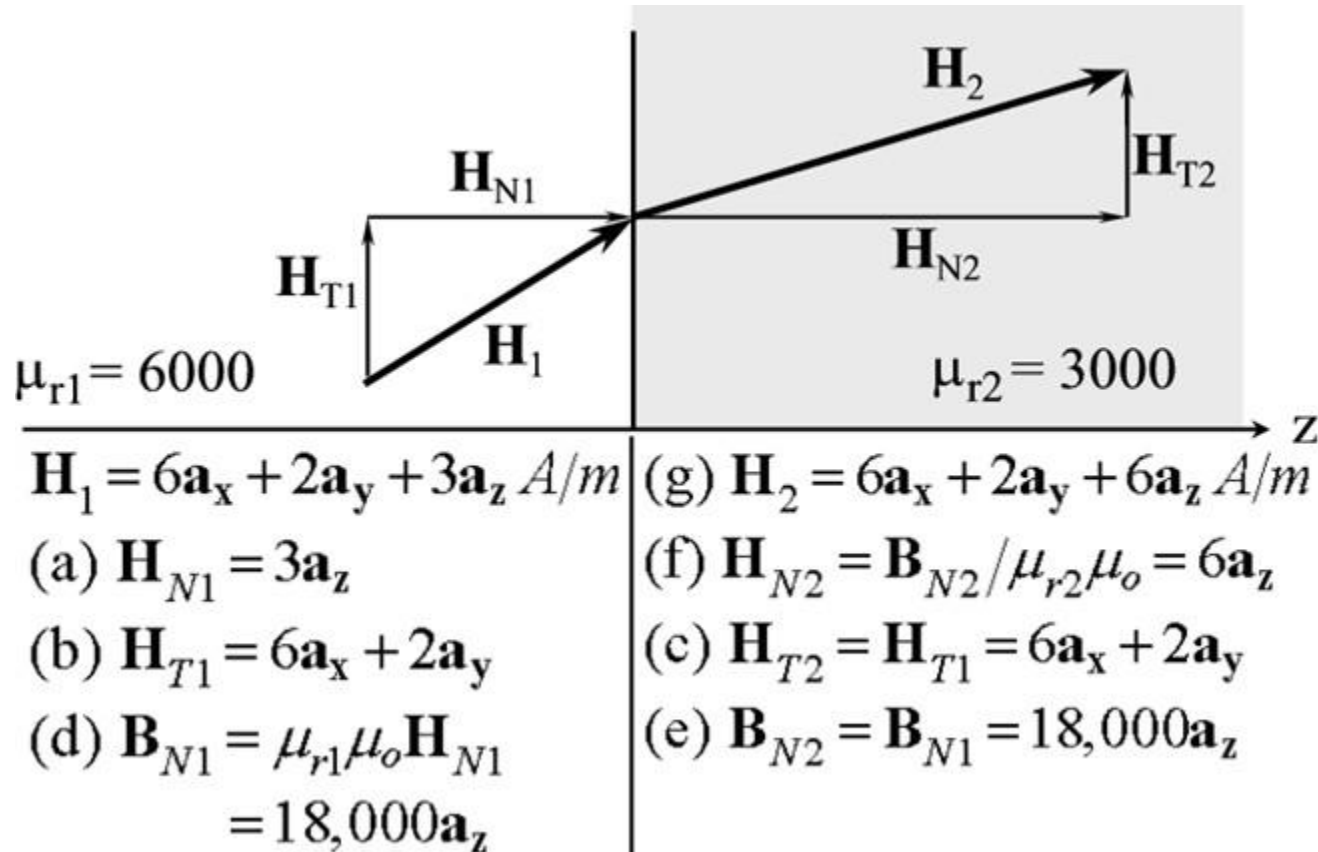


The tangential component of the magnetic flux density $\mathbf{B}$ is not continuous across the boundary.

Example



The magnetic field intensity is given as $\mathbf{H}_1 = 6\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$ (A/m) in a medium with $\mu_{r1} = 6000$ that exists for $z < 0$. We want to find $\mathbf{H}_2$ in a medium with $\mu_{r2} = 3000$ for $z > 0$.

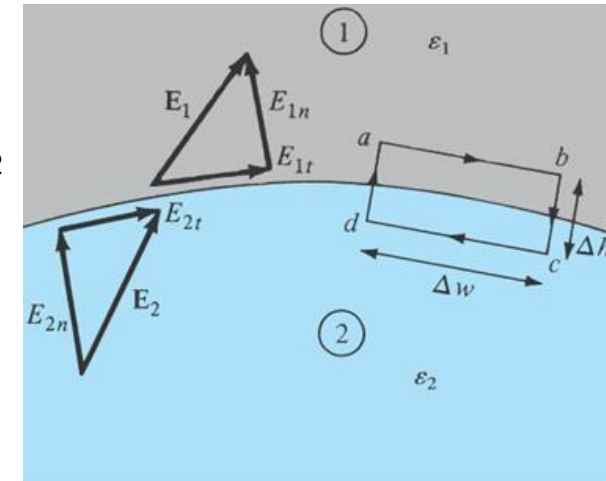


Boundary Conditions For Electrostatic Fields



Two Dielectric media:

- E_t is **continuous** across the boundary. $E_{t1} = E_{t2}$
- D_t is **discontinuous** across the boundary.



Helpful in determining how the E and D vectors change in crossing an interface.

Boundary Conditions For Electrostatic Fields

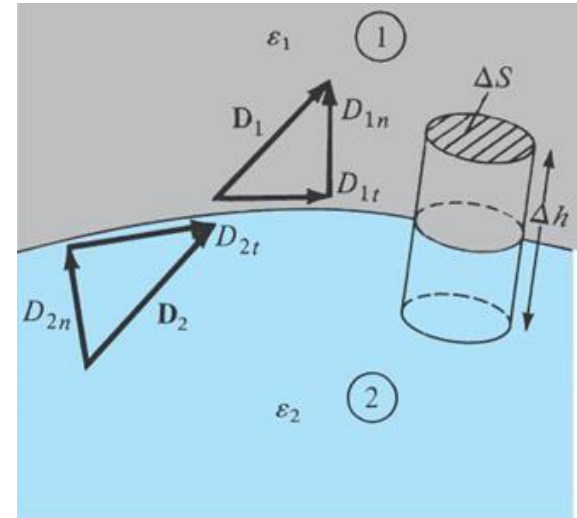


$$D_{n1} - D_{n2} = \rho_s$$

E_n is **discontinuous** across the charge free boundary.

$$[D_{n1} - D_{n2} = \rho_s]$$

$$[E_{t1} = E_{t2}]$$



Boundary Conditions: Special Case



Perfect
Dielectric
(1)

Perfect
Dielectric
(2)

$$\frac{\sigma = 0}{\rho_s = 0}$$
$$\int_0 J = 0$$

$$\begin{aligned} B_{N1} &= B_{N2} \\ H_{t1} &= H_{t2} \\ D_{N1} &= D_{N2} \\ E_{t1} &= E_{t2} \end{aligned}$$

Boundary Conditions: Special Case



Perfect
Dielectric

①

Perfect
Conductor

②

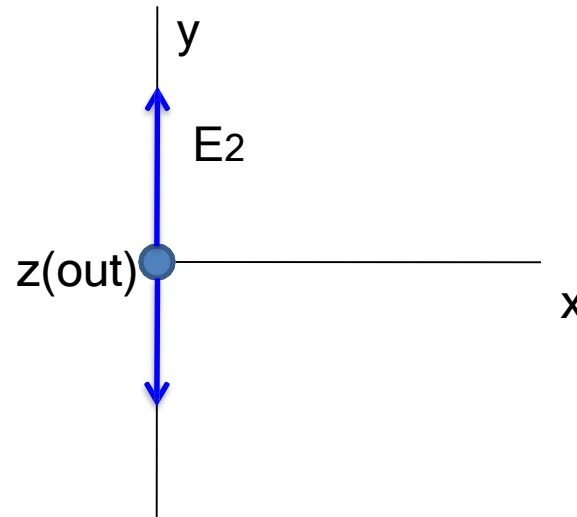
→ In conductor
field is
zero

$$\left[\begin{array}{l} B_{N1} = 0 \\ H_{t1} = J_s \\ D_{N1} = \rho_s \\ E_{t1} = 0 \end{array} \right]$$

Polarization of Plane Waves



Wave polarization describes the **shape** and **locus of tip** of the vector $\mathbf{E}$ at a given point in space as a function of time.



Polarization of a uniform plane wave describes the time-varying behaviour of the electric field intensity vector at a given point in space.

Polarization of Plane Waves



A uniform plane wave traveling in the +z direction may have x- and y-components.

$$\mathbf{E}(z) = \hat{a}_x E_x(z) + \hat{a}_y E_y(z)$$

$$E_x = E_1 \sin(\omega t - \beta z)$$

$$E_y = E_2 \sin(\omega t - \beta z + \delta)$$

at $z=0$

$$E_y = E_2 \sin \omega t \cos \delta + E_2 \cos \omega t \sin \delta$$

$$\frac{E_y^2}{E_2^2} - \frac{2E_x E_y}{E_1 E_2} \cos \delta + \frac{E_x^2}{E_1^2} = \sin^2 \delta$$

Polarization of Plane Waves

- A uniform plane wave traveling in the +z direction may have x- and y- components.

$$\mathbf{E}(z) = \hat{\mathbf{a}}_x E_x(z) + \hat{\mathbf{a}}_y E_y(z)$$

$$E_x = E_1 \sin(\omega t - \beta z)$$

$$E_y = E_2 \sin(\omega t - \beta z + \delta)$$

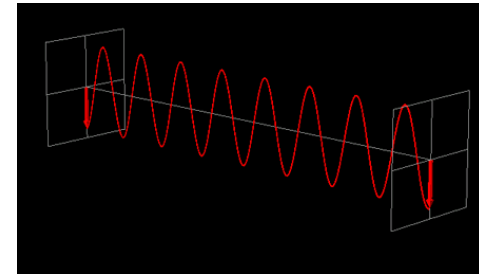


$$\frac{E_y^2}{E_2^2} - \frac{2E_x E_y}{E_1 E_2} \cos \delta + \frac{E_x^2}{E_1^2} = \sin^2 \delta$$

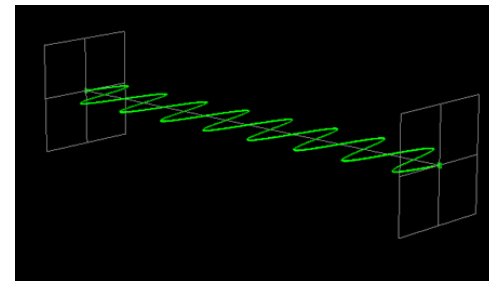
Generally an Elliptically polarized wave

$$\frac{E_y^2}{E_2^2} - \frac{2E_x E_y}{E_1 E_2} + \frac{E_x^2}{E_1^2} = 0$$

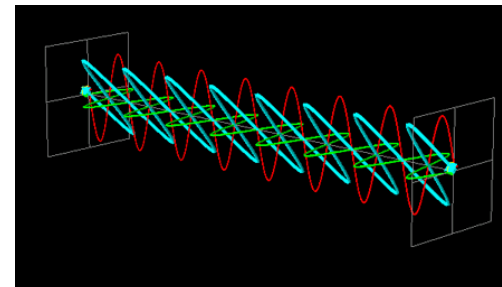
$$\delta = 0$$



$$E_1 = 0$$



$$E_2 = 0$$

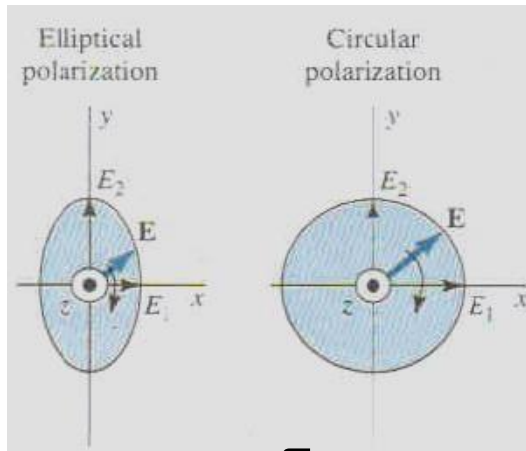


$$\delta = 0$$

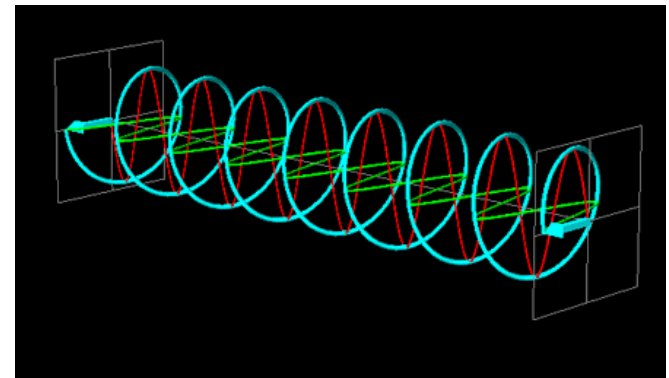
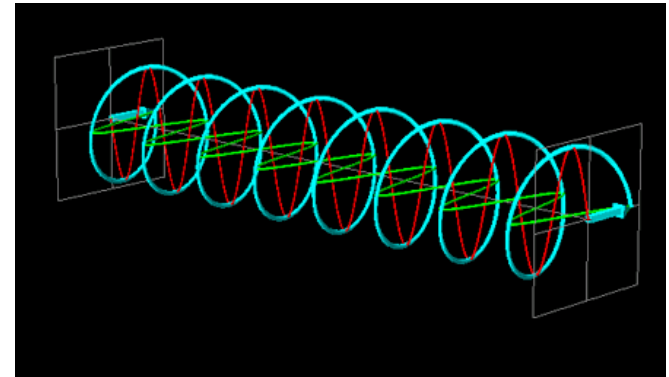
Polarization of Plane Waves



$$\frac{E_y^2}{E_2^2} - \frac{2E_x E_y}{E_1 E_2} \cos \delta + \frac{E_x^2}{E_1^2} = \sin^2 \delta$$



$$E_1 \neq E_2 \quad \delta = \pm \frac{\pi}{2} \quad E_1 = E_2$$

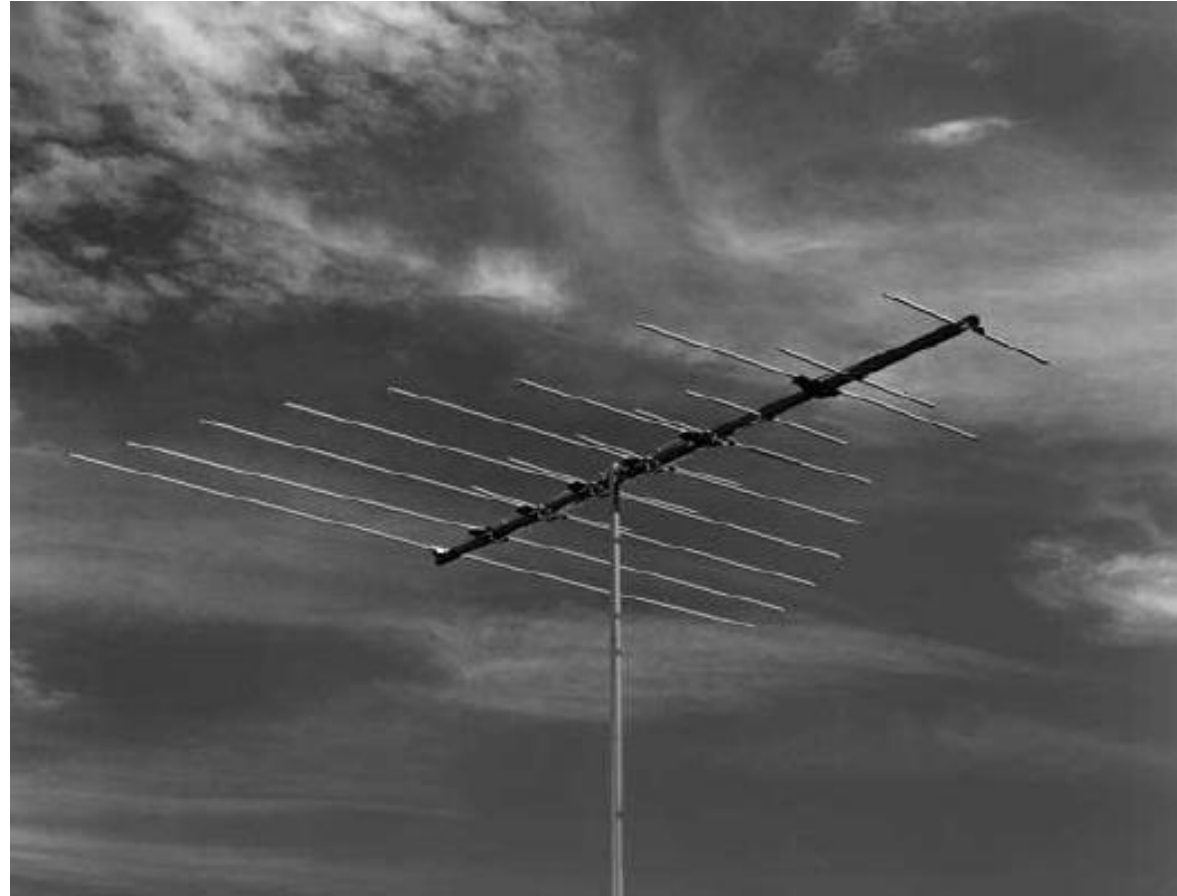


Polarization

Practical application



AM broadcast antenna



Roof top TV receiving antenna

Example



Determine the polarization of the wave if the electric field is given by

$$\vec{E} = (4\hat{a}_x - j2\hat{a}_y)e^{-j0.5z}$$